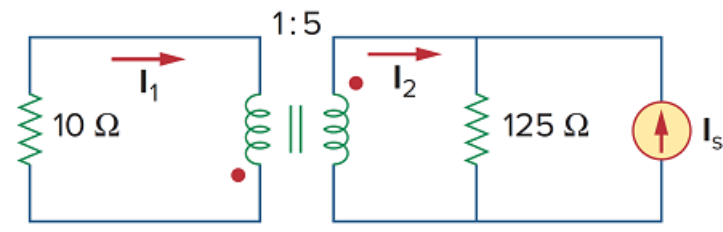


Problem

Given $I_2 = 2\text{ A}$, determine the value of I_s in Fig. 13.106.

Figure 13.106



Step-by-step solution

Step 1 of 3

Refer to Figure 13.106 in the textbook for the actual circuit.

Redraw the circuit as follows.

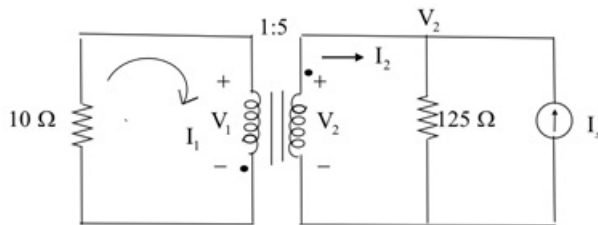


Figure 1

Step 2 of 3

Apply Kirchhoff's voltage law to mesh 1.

$$10I_1 + V_1 = 0 \quad \dots\dots (1)$$

At the transformer terminals,

$$\frac{I_1}{I_2} = -\frac{N_2}{N_1}$$

Substitute 5 for N_2 , 1 for N_1 .

$$\frac{I_1}{I_2} = -5$$

$$I_1 = -5I_2$$

Substitute 2 A for I_2 .

$$\begin{aligned} I_1 &= -5(2) \\ &= -10\text{ A} \end{aligned}$$

Substitute -10 A for I_1 in equation (1).

$$10(-10) + V_1 = 0$$

$$V_1 = 100\text{ V}$$

Step 3 of 3

Apply Kirchhoff's current law at secondary side.

$$I_2 + I_s = \frac{V_2}{125}$$

$$I_s = \frac{V_2}{125} - I_2 \dots\dots (2)$$

At the transformer terminals,

$$\frac{V_2}{V_1} = -\frac{N_2}{N_1}$$

Substitute 5 for N_2 , 1 for N_1 .

$$\frac{V_2}{V_1} = -5$$

$$V_2 = -5V_1$$

Substitute 100 V for V_1 .

$$\begin{aligned} V_2 &= -5(100) \\ &= -500 \text{ V} \end{aligned}$$

Substitute -500 V for V_2 and 2 A for I_2 in equation (2).

$$\begin{aligned} I_s &= \frac{-500}{125} - 2 \\ &= -6 \text{ A} \end{aligned}$$

Therefore, the value of I_s is -6 A.